

Lab 1 - Requirements Table | PS #36: Restaurant Table Booking & Pre-Ordering App

Actors: Diner, Restaurant Manager, Payment Gateway (external), Kitchen Display System (external)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a diner to select an open table from an interactive 2D floor plan and link pre-ordered menu items to that reservation. [GIVEN]	High	Pass: a table marked "open" is reserved and a kitchen pre-order ticket is queued against the reservation. Fail: the same table is double-booked for an overlapping dining window.	Core booking-and-pre-order functionality of the app.
FR-002	Functional	The system shall compute an estimated kitchen preparation timeline from the diner's pre-ordered items and reservation arrival time, and display the target food-ready time.	High	Pass: for a given pre-order and arrival time, the displayed food-ready time matches arrival time within +/-5 minutes in >=90% of test orders. Fail: no timeline shown, or deviation > 15 minutes.	The "seamless arrival dining" promise depends on accurate prep timing.
FR-003	Functional	The system shall let a diner pay a deposit or full amount for the reservation and pre-order through an integrated payment gateway before the booking is confirmed.	High	Pass: on gateway authorization the reservation status becomes "Confirmed" and a receipt is issued. Fail: a reservation is marked Confirmed without a successful authorization.	Secures revenue and reduces no-shows on pre-ordered meals.
FR-004	Functional	The system shall allow the Restaurant Manager to change a table's status (Available -> Reserved -> Seated -> Billed) from any floor terminal.	Medium	Pass: a status change made on one terminal is reflected on the floor plan and on all other terminals. Fail: a stale status persists after a change.	Managers need real-time floor control to avoid seating conflicts.
FR-005	Functional	The system shall send the diner a booking confirmation containing reservation date/time, table number and pre-order summary via app notification and email/SMS.	Medium	Pass: a confirmation containing all three data fields is delivered within 30 s of booking. Fail: any field missing, or no delivery within 30 s.	Confirms the booking and gives the diner an arrival reference.
NFR-001	Non-functional (Performance)	Table status changes (Reserved, Seated, Billed) shall sync across all manager floor terminals in under 500 ms. [GIVEN]	High	Pass: benchmarking under simulated peak load confirms 95th-percentile sync latency < 500 ms. Fail: latency >= 500 ms.	Prevents overbooking and keeps every floor terminal consistent.
NFR-002	Non-functional (Security)	The system shall transmit and store all payment and personal data over encrypted channels (TLS 1.2+) and handle card data in compliance with PCI-DSS.	High	Pass: a security scan finds no plaintext PII or card data in transit or at rest and confirms TLS 1.2+. Fail: any unencrypted PII or card data detected.	Protects diner data and meets legal / industry obligations.

Legend:

[GIVEN] = requirement supplied in the problem statement
Functional requirement
Non-functional requirement